When Visibility Shapes Timing:
Evidence on Bidding Dynamics and Market Efficiency

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Abstract

Online service procurement platforms must help clients identify qualified providers quickly while preserving effective competition and match quality. A central design choice is whether providers can observe competing bids during the procurement process. Although transparency can facilitate learning and price discovery, it may also create an operational cost by encouraging providers to delay participation while monitoring competitors. We study a platform-wide transition from open to sealed bidding in a large online labor market, a setting characterized by dynamic worker arrival, multi-attribute client selection, and client-controlled early stopping. We argue that bid visibility creates a transparency-timing tradeoff: open bidding increases the informational value of waiting, whereas sealed bidding reduces this value while preserving the risk that opportunities disappear before delayed bids are submitted. Using data from three months before and after the policy change, we find that workers submit bids significantly earlier when competing bids are hidden. Sealed bidding is associated with faster contract awards and a higher probability of contract formation. These market-clearing gains occur despite lower bidder participation and lower average submitted bid amounts, suggesting that competition is restructured rather than weakened. Supplemental analyses show that sealed bidding also increases the early arrival of bids, including bids from experienced workers. We also find higher employer ratings after the transition to sealed bidding. Our findings identify bid timing as an important mechanism through which information visibility affects procurement outcomes and show that transparency policies should be evaluated as operational process-design choices in online service procurement.

Keywords: information visibility; online service procurement; bid timing; market-clearing efficiency; online labor markets

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1. Introduction

Online service procurement platforms must solve a difficult operating problem: they need to help buyers identify qualified service providers quickly while preserving enough competition and information for good matching decisions. A central design choice in these markets is whether competing bids should be visible during the procurement process. Transparency is often viewed as beneficial because it allows market participants to learn from others, reduces uncertainty, and improves decision-making (Milgrom and Weber 1982; Arora et al. 2007; Greenwald et al. 2010; Lu et al. 2019). Yet transparency can also create a hidden operational cost. When competing bids are visible, service providers may postpone participation to observe competitors before committing to an offer. In many online service procurement settings, however, buyers are not required to wait until a posted deadline and may award a contract as soon as a satisfactory provider appears. Information that helps service providers learn may therefore slow the arrival of actionable offers to buyers. This creates a fundamental tension: bid visibility may improve information while reducing the speed with which markets clear.

Online labor markets provide a particularly useful context for examining this tension. In these markets, employers use platform-based procurement mechanisms to hire geographically dispersed workers for customized projects (Gefen and Carmel 2008; Agrawal et al. 2013; Lin et al. 2018; Guo et al. 2026). The procurement process is buyer-determined rather than purely price-based: employers evaluate proposed wages together with nonprice attributes such as worker reputation, prior experience, location, and expected service quality. At the same time, workers arrive asynchronously, discover projects at different times, and decide whether and when to bid. Importantly, employers can stop the process before bidding formally concludes by awarding the contract once a satisfactory worker appears (Liang et al. 2022; Bai et al. 2026). These features make online labor markets a useful setting for studying the operational consequences of bid visibility because bid visibility operates in a setting where information acquisition, worker arrival, and buyer-controlled stopping are tightly intertwined.

More broadly, existing research has primarily examined the outcomes of information visibility, including participation, prices, bidder learning, allocation, and buyer surplus (Arora et al. 2007; Greenwald et al. 2010; Kannan 2012; Haruvy and Katok 2013; Hong et al. 2016; Lu et al. 2019). Much less attention has been devoted to the bidding process itself, particularly how visibility influences the timing of bid submission and how bid timing affects market-clearing efficiency. This omission matters because the timing of bids is an operationally consequential feature of service procurement. In online labor markets, a worker who delays participation to gather information may submit a better-informed bid, but the delay may also slow employers' access to viable candidates or cause the worker to miss the opportunity altogether. Consequently, the performance effects of bid visibility may arise not only through what workers bid, but also through when they bid.

We develop a process-based explanation in which bid visibility changes the value of waiting and, through bid timing, affects the speed and likelihood of market clearing. Under open bidding, workers can observe competing bids and bidder characteristics, creating incentives to wait and learn before submitting an offer. Under sealed bidding, the informational benefit of waiting is reduced, while the risk that the employer may select another worker remains. This shift should lead workers to bid earlier. Earlier bid arrival, in turn, should increase employers' access to actionable offers, enabling faster contract awards and increasing the probability that posted projects result in contracts. The same mechanism may also affect downstream match quality: if viable offers arrive earlier, employers may have greater opportunity to screen workers before selecting a match rather than making decisions after prolonged waiting or limited early participation.

We examine these arguments using a platform-wide transition from open to sealed bidding, which we refer to as the "regime change," in a large online labor market. We find that workers submit bids significantly earlier after competing bids become hidden. Employers subsequently award contracts more quickly, and posted projects are more likely to result in contract formation. Supplementary project-level analyses further show that sealed bidding accelerates the early arrival of offers, including offers from experienced workers. These market-clearing improvements occur even though fewer workers participate in each project. We also find that average bid amounts decline after the policy change, suggesting that sealed bidding restructures competition rather than simply weakening it. We further find higher employer ratings for completed projects after the visibility change. Taken together, the findings suggest that reducing bid visibility can accelerate market clearing without reducing observed post-award match quality.

This study makes three major contributions. First, we contribute to research on information disclosure and procurement market design by identifying bid timing as an important mechanism through which visibility affects procurement outcomes (Arora et al. 2007; Greenwald et al. 2010; Kannan 2012; Haruvy and Katok 2013; Lu et al. 2019). Prior work has largely evaluated transparency through outcomes such as prices, participation, learning, and surplus. We show that visibility also affects the temporal flow of offers, which is critical when buyers can stop the procurement process early. Second, we contribute to research on bid timing by showing that information visibility shapes incentives to delay participation and thereby affects market-clearing performance (Roth and Ockenfels 2002; Ockenfels and Roth 2006; Ely and Hossain 2009). Third, we contribute to the literature on online labor markets and service procurement by identifying dynamic worker arrival and employer-controlled stopping as operational boundary conditions that shape the value of transparency (Hong et al. 2016; Liang et al. 2023; Bai et al. 2026). Our findings suggest that transparency is not universally beneficial or harmful; its value depends on whether the benefits of bidder learning outweigh the operational costs of delayed participation.

The remainder of the paper proceeds as follows. Section 2 reviews related literature. Section 3 presents the institutional setting and hypotheses. Section 4 describes the data and variables. Section 5 examines the effect of bid visibility on bid timing. Section 6 reports the effect of bid visibility on market-clearing outcomes. Section 7 examines the impact of bid visibility on competitive tradeoffs and post-award match quality. Section 8 discusses theoretical contributions, managerial implications, limitations, and directions for future research.

2. Related Literature

We position our study at the intersection of online service procurement, information visibility, and bid timing. Together, these streams help explain how bid visibility shapes when workers submit offers and how this timing affects market-clearing efficiency and match quality.

2.1 Buyer-Determined Online Service Procurement

Online labor markets are a form of digital service procurement in which employers post customized projects, workers submit offers, and employers select service providers based on both price and nonprice attributes. These markets differ from standard price-only procurement because the service being procured is often customized, worker quality is uncertain before delivery, and employers retain substantial discretion over whom to hire. Prior research shows that online labor markets reduce geographic barriers and allow employers to access geographically dispersed service providers, but they also create information frictions because employers must evaluate workers they may not know personally (Gefen and Carmel 2008; Agrawal et al. 2013; Stanton and Thomas 2015; Guo et al. 2026).

In buyer-determined service procurement, the lowest price does not necessarily win. Employers evaluate proposed wages together with worker reputation, prior experience, location, communication, and other signals of expected service quality. This logic is consistent with multidimensional procurement theory, which emphasizes that buyers jointly evaluate price and quality when selecting suppliers (Che 1993; Asker and Cantillon 2008). It is also consistent with research on beauty-contest auctions, in which bidders compete on multiple dimensions and the buyer's selection rule is only partially observable to participants (Yoganarasimhan 2016). Thus, online labor markets are better understood as buyer-determined service procurement systems than as simple price-based auctions.

This procurement process creates uncertainty for both sides of the market. Employers must infer worker quality before the service is delivered, while workers must decide whether, when, and how to bid without fully knowing how employers will evaluate competing price-quality combinations. Prior research shows that reputation systems, agent affiliation, platform-provided signals, and prior contracting histories help reduce these information asymmetries and facilitate contracting in online service markets (Moreno and Terwiesch 2014; Pallais 2014; Kokkodis and Ipeirotis 2015; Hong and Pavlou 2017; Lin et al. 2018; Horton 2019; Barach et al. 2020; Burtch et al. 2021; Gu and Zhu 2021; Hong and Shao 2021; Hong et al. 2021; Kokkodis 2021, 2023; Troncoso and Luo 2023). More recent work further shows that quality signals do not always have uniformly positive effects. Certification can improve employer screening but may also alter worker participation when obtaining the signal is costly (Bai et al. 2026), and novel signaling mechanisms may backfire when employers interpret them as negative rather than positive indicators of worker quality (Gao et al. 2023). These studies highlight that platform information design affects not only what employers know about workers, but also how workers choose to participate, signal quality, and compete for projects.

Although this literature has substantially advanced understanding of online service procurement, it has focused primarily on selection, participation, reputation, signaling, and contracting outcomes. Less attention has been devoted to the temporal operating process through which offers arrive and employers decide when to stop searching. This omission is important because, in buyer-determined service procurement, information affects not only how employers evaluate workers but also how workers decide when to participate. Our study contributes to this stream by showing that platform information design can shape market-clearing efficiency through its effect on the timing of worker participation. Bid visibility is especially important in this process because it determines whether workers can observe competing offers before submitting their own. We therefore next review research on information visibility in procurement markets.

2.2 Information Visibility in Procurement Markets

Bid visibility is a central feature of procurement market design because it determines the extent to which bidders can observe competitors' offers during the bidding process. Prior research has examined how alternative disclosure policies influence bidder learning, competitive behavior, and procurement outcomes. Information disclosure can facilitate learning and price discovery by providing bidders with additional information about the competitive environment (Milgrom and Weber 1982; Arora et al. 2007; Greenwald et al. 2010). At the same time, disclosure may also change bidders' strategic behavior, and its effects depend on the type of uncertainty bidders face and the information being revealed.

Prior work identifies several channels through which information revelation affects procurement outcomes. Arora et al. (2007) show that the effectiveness of disclosure policies depends on uncertainty about market structure. Greenwald et al. (2010) examine the performance of alternative disclosure policies across repeated procurement sessions, while Kannan (2012) demonstrates that bid-revelation rules influence bidder learning under cost uncertainty. Elmaghraby et al. (2012) show that rank feedback shapes bidding behavior in procurement auctions. Haruvy and Katok (2013) find that reducing information can increase buyer surplus in multi-attribute procurement settings. Lu et al. (2019) show that winner-identity disclosure affects bidder learning and future participation decisions. Related work on buyer-determined auctions shows that transparency regarding quality information can affect bidder behavior and procurement outcomes (Stoll and Zöttl 2017). More recent work on online service operations similarly shows that information transparency can shape matching efficiency, congestion, and platform welfare, suggesting that transparency policies should be evaluated not only as information-disclosure choices but also as operational design choices (Xu et al. 2024).

The study most closely related to ours is Hong et al. (2016), which compares open and sealed bidding formats in an online labor market. Hong et al. develop a tradeoff between valuation uncertainty and competition uncertainty and show that bid visibility affects participation, buyer surplus, and contracting outcomes. Although Hong et al. (2016) is closely related to our setting, the two studies examine fundamentally different questions. Hong et al. treat bid visibility as a buyer-level design choice and focus on how visibility formats affect bidder uncertainty, participation, and buyer surplus when buyers choose different visibility formats. In contrast, we examine a platform-level change in bid visibility rules that affects all market participants simultaneously and focus on how information visibility influences bid timing and subsequent contracting outcomes.

Taken together, this literature establishes that information visibility affects procurement outcomes, but it has paid less attention to the temporal process through which offers become available to buyers. Most studies evaluate disclosure policies through outcomes such as prices, participation, surplus, or allocation efficiency. In buyer-determined online service procurement, however, the timing of bid arrival is itself operationally consequential because buyers may stop the process before the posted deadline. A visibility policy that helps workers learn from competitors may also encourage waiting, monitoring, and bid revision, thereby slowing the arrival of actionable offers. Thus, bid visibility is not only an information-disclosure policy; it is also a process-design choice that may shape the temporal flow of bids available to buyers. To develop this timing-based mechanism, we next turn to research on bid timing and market-clearing dynamics.

2.3 Timing in Procurement Markets

Research on bid timing provides a useful foundation for understanding how market rules shape when participants act. Studies of Internet auctions document systematic patterns of delayed participation and show that market rules influence when bidders choose to submit bids (Roth and Ockenfels 2002;

Ockenfels and Roth 2006; Ely and Hossain 2009). Other work demonstrates that transaction costs, waiting costs, and auction mechanisms affect bid dynamics and market outcomes (Carare and Rothkopf 2005; Katok and Kwasnica 2008; Shachat and Wei 2012). More broadly, research on market timing shows that institutional rules can shape when participants enter, search, and transact (Roth and Xing 1994).

Although this literature highlights the importance of timing, much of it focuses on fixed ending rules, last-minute bidding, Dutch auction clocks, transaction costs, or alternative auction mechanisms. Less attention has been devoted to timing in buyer-determined service procurement, where bidders arrive over time, and buyers retain discretion over when to stop evaluating offers. This distinction is important because procurement performance may depend not only on the final set of bids, but also on the temporal availability of acceptable offers.

We bring this timing perspective together with research on information visibility. The information-disclosure literature explains how visibility affects learning, competition, and surplus, while the timing literature explains how market rules affect when participants act. We integrate these perspectives by examining whether and how information visibility changes the timing of bid submission and, through this timing channel, affects market-clearing efficiency in online service procurement.

To develop this argument, we next examine the institutional features of our studied online labor platform and explain why bid timing becomes a central strategic decision under different visibility regimes.

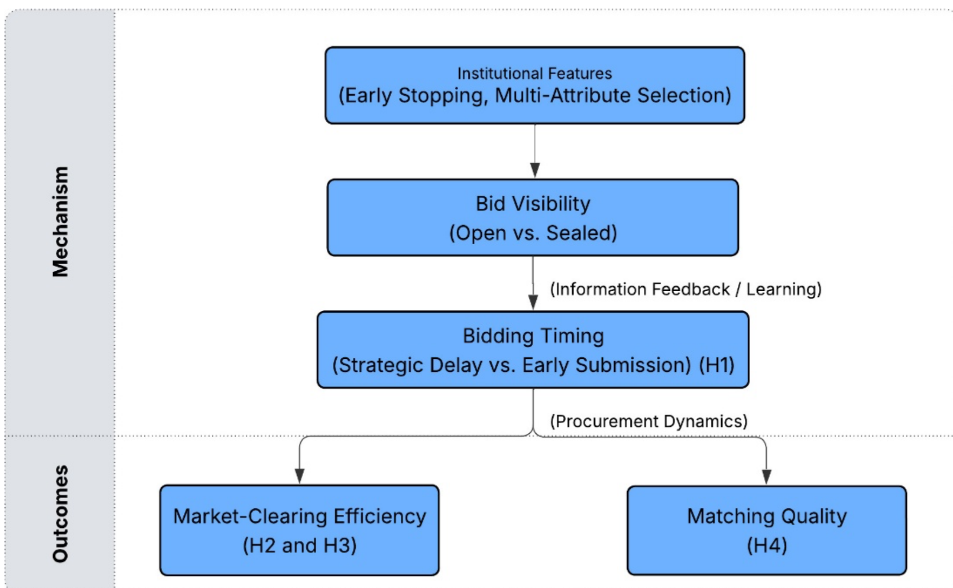
3. Theoretical Development

3.1 Institutional Setting and Key Features

Our theoretical framework builds on three common institutional features of online labor markets: employer-controlled early stopping, dynamic worker arrival, and bid visibility. Together, these features make bid timing a strategically important decision because workers must balance the informational benefits of waiting against the risk of losing an opportunity. First, employers are not required to wait until a predetermined deadline before making a hiring decision. Instead, employers may terminate the procurement process at any time by selecting a worker or canceling the project altogether. Such employer-controlled stopping differs substantially from the fixed-deadline settings commonly examined in studies of bidding behavior. Second, workers arrive dynamically over time rather than simultaneously. Similar to many online marketplaces, project opportunities become visible to workers asynchronously, and workers discover, evaluate, and respond to projects at different points in time. Third, the visibility of competing bids differs across bidding formats. Under open bidding, workers can observe previously submitted bids and information about competing workers, whereas such information remains hidden under sealed bidding. Consequently, workers face different informational environments when making bidding decisions.

These institutional features make bid timing a central strategic decision in online labor markets. Figure 1 summarizes the conceptual framework. Dynamic worker arrival and employer-controlled early stopping create incentives for workers to bid promptly to avoid losing opportunities, whereas information visibility may increase the value of waiting by allowing workers to observe and learn from competitors. Thus, bid visibility creates a tradeoff between the informational benefit of waiting and the operational cost of delayed participation. The interaction of these competing forces influences bid timing, which subsequently affects market-clearing outcomes and post-award match quality. We develop these arguments below.

Figure 1. Conceptual Framework: Information Visibility, Bid Timing, and Market Outcomes



3.2 Visibility and Workers' Timing Decisions

Workers in online labor markets decide not only whether to bid and what wage to propose, but also when to submit a bid. Because workers arrive asynchronously and employers may award contracts before the posted deadline, bid timing becomes an important strategic choice.

Under open bidding, workers can observe previously submitted bids and information about competing workers. Because workers arrive and bid over time, information about the competitive environment accumulates throughout the life of a project. Delaying participation may therefore allow workers to observe additional bids and learn about the intensity and nature of competition before committing to an offer. These observations provide information about rival bidders and prevailing competitive conditions. Prior research suggests that bidders often delay participation when waiting allows them to learn from competitors or avoid revealing information prematurely (Roth and Ockenfels 2002). At the same time, submitting a bid early discloses information that later-arriving competitors may observe. Consequently, open bidding creates an informational benefit to waiting.

However, waiting is not without cost. Prior research shows that bidders respond strategically to auction-ending rules and the possibility that opportunities disappear while they wait (Ely and Hossain 2009; Ockenfels and Roth 2006). In our setting, a worker who delays participation risks losing the opportunity to win if the employer selects another worker before the stated deadline. Workers therefore face a tradeoff between acquiring additional information and preserving the opportunity to compete.

Under sealed bidding, the informational benefit of waiting is substantially reduced because competing bids are not observable. Although the risk of exclusion remains, workers receive little additional information from delaying participation. As a result, the balance shifts toward earlier bidding. Therefore, we hypothesize that:

Hypothesis 1. *Conditional on submitting a bid, workers submit their bids earlier under sealed bidding than under open bidding.*

3.3 Bid Timing and Market-Clearing Efficiency

The timing of bid arrival influences not only worker behavior but also the efficiency with which employers complete the procurement process. In online labor markets, employers are free to award a project at any point during the bidding process. As a result, the timing at which acceptable candidates become available directly affects both how quickly and how successfully employers can complete a match.

In buyer-determined online labor markets, employers evaluate both price and nonprice attributes when selecting a worker. This multi-attribute selection logic is consistent with procurement theory, which shows that buyers often evaluate suppliers using both price and nonprice dimensions (Che 1993; Asker and Cantillon 2008), and with empirical research on online labor markets showing that employers consider worker reputation, experience, and other quality signals when making hiring decisions (Moreno and Terwiesch 2014; Hong and Pavlou 2017; Lin et al. 2018). Earlier arrival of viable price-quality combinations allows employers to begin screening and evaluation sooner. It increases the likelihood that an acceptable worker becomes available while the employer remains actively engaged with the project. In contrast, strategic delay under open bidding may slow the accumulation of actionable offers. When workers postpone participation to observe competitors and gather information, employers receive fewer offers during the early stages of the procurement process. As a result, employers may need to wait longer before observing a satisfactory set of potential workers.

This logic suggests two distinct dimensions of market-clearing efficiency. The first is the speed of market clearing among projects that ultimately result in a contract. If sealed bidding reduces workers' incentives to delay participation, acceptable offers should become available sooner, allowing employers to evaluate candidates and award contracts more quickly. Therefore, we hypothesize:

Hypothesis 2. *Conditional on contract formation, employers award contracts sooner under sealed bidding than under open bidding.*

The second dimension is the probability of market clearing. Because employers may terminate the procurement process before the posted deadline, delayed bid arrival can increase the likelihood that a project remains unfilled or is abandoned before a satisfactory worker is identified. Earlier arrival of viable offers should reduce this risk by increasing the probability that employers encounter acceptable candidates during the active evaluation window. Therefore, we hypothesize:

Hypothesis 3*. Posted projects are more likely to result in contract formation under sealed bidding than under open bidding.*

3.4 Bid Timing and Match Quality

The preceding arguments focus on whether sealed bidding accelerates market clearing and increases the likelihood of contract formation. An important remaining question is whether these operational gains come at the expense of match quality. Faster contracting could potentially reduce match quality if employers make decisions with less information, face a smaller pool of candidates, or rush to select workers before adequately evaluating their price and nonprice attributes. This concern is particularly relevant in online service procurement because service quality is difficult to assess before delivery, and employers rely on imperfect signals such as worker reputation, prior experience, location, communication, prior performance, and platform-provided quality indicators when selecting among competing workers (Hong and Pavlou 2017; Lin et al. 2018; Hong et al. 2021; Kokkodis and Ransbotham 2023; Bai et al. 2026).

However, sealed bidding may also improve post-award match outcomes through its effect on bid timing. In buyer-determined service procurement, employers do not simply choose the lowest bid; rather, they compare multiple price-quality combinations and select the worker they perceive as most suitable for the project based on a combination of factors. When viable candidates arrive earlier, employers have more time to screen workers, compare alternatives, communicate if needed, and select a suitable match before attention to the project declines. Earlier bid arrival may therefore improve the employer's opportunity to evaluate candidates rather than forcing a decision after prolonged waiting or limited early participation.

Sealed bidding may further improve match quality by reducing strategic monitoring and repeated bid adjustment. Under open bidding, workers can observe competitors and revise their behavior in response to visible information. Such behavior may increase bidding activity, but it does not necessarily increase the quality of the eventual match. Under sealed bidding, workers must submit offers without observing rivals' bids, which may encourage more deliberate initial bids from workers who are genuinely interested in the project. As a result, the candidate pool may become smaller but more immediately actionable, allowing employers to identify suitable workers more efficiently.

Taken together, these arguments suggest that sealed bidding need not create a tradeoff between faster market clearing and poorer match quality. Instead, by accelerating the arrival of viable offers and preserving employers' ability to conduct multi-attribute screening, sealed bidding may lead to better observed post-award match outcomes among contracted projects. Therefore, we hypothesize:

Hypothesis 4. *Conditional on contract formation, contracts formed under sealed bidding generate better observed post-award match outcomes than contracts formed under open bidding.*

4. Data and Variables

4.1 Data

Our research focuses on an online labor market headquartered in the United States, where employers and workers from around the world engage in project-based service transactions.[1] The platform hosts a variety of projects, with categories such as website and software development, graphical design, data entry and management, and writing. We have access to comprehensive project, employer, and worker data through collaboration with the platform. Specifically, we have detailed information about each project (e.g., when it is posted and closed), each employer (e.g., registration history and location), each worker (e.g., reputation, past bidding, winning, and performance history), each bid (e.g., bid amount and bid time), project outcome (e.g., the selected worker), and post-award outcome (e.g., rating given by employers).

The platform originally operated using the open bidding format but transitioned abruptly to the sealed bidding format on a single implementation date without prior notice to participants. This platform-wide

[1] We cannot disclose the platform and other details (e.g., dates and exact platform feature names) that may reveal this platform due to a non-disclosure agreement.

policy change provides a useful quasi-experimental setting for examining the impact of bid visibility, while maintaining consistency in other platform operations shortly before and after the regime change. We collected data on all projects and their bids posted three months before and after this regime change. During this narrow timeframe, we are not aware of other major platform mechanisms or policies introduced, and our composition tests show no systematic discontinuities in observable employer, worker, or project characteristics around the visibility change. The final dataset comprises 1,926 projects launched by 967 employers; 3,421 workers placed 16,581 bids; among these projects, 802 resulted in contracts. Our empirical analyses are structured into two levels: bid-level analyses examining bidding behavior and project-level analyses examining market-clearing, competition, and post-award outcomes.

4.2 Variables

Dependent Variables. Consistent with our theoretical framework, we examine three groups of primary outcome variables: bid timing, market-clearing efficiency, and post-award match quality. We also examine participation and average bid amounts as supplementary outcomes that capture changes in the competitive structure of the bidding process.

Bid Timing (H1). To examine whether changes in bid visibility affect workers' bidding behavior, we construct *BidTime*, measured as the elapsed time (in hours) between project posting and bid submission. Because the distribution is highly skewed, we use its logarithmic transformation in the empirical analyses.

Market-Clearing Efficiency (H2 and H3). We use two project-level measures of market-clearing efficiency. *TimeToAccept* measures the elapsed time between project posting and the employer's selection of a winning worker. This variable captures the speed of market clearing and is observed only for projects that result in a contract award. *ProjectSuccess* is an indicator variable equal to one if the project results in a contract award and zero otherwise. This variable captures the probability of market clearing.

Match Quality (H4). To examine whether changes in match quality accompany changes in market-clearing efficiency, we use two post-award outcomes. *WorkerRating* captures employer satisfaction by measuring the rating assigned to the winning worker upon project completion. *Rehire* is an indicator variable equal to one if the employer subsequently hires the same worker for a future project.

Supplementary Outcomes: Participation and average bid amounts. To examine how bid visibility affects the competitive structure of the bidding process, we also consider two supplementary outcomes. *#OfBidders* measures the number of unique workers participating in a project and captures realized worker participation. *AvgBidAmount* measures the average wage proposed by participating workers in a project and captures submitted price competition. Lower average bid amounts indicate more aggressive submitted bids among participating workers.

Independent Variable. The primary independent variable is *AfterChange*, an indicator variable equal to one if the project was posted after the platform changed from open bidding to sealed bidding, and zero otherwise.

Control Variables. We include four groups of control variables in our analyses: (1) Employer characteristics such as employer's prior project experience (*EmployerExperience*) and geographic region (*Region*); (2) Worker characteristics such as prior project experience (*WorkerExperience*), whether the worker is a new worker (*NoRating*), and the number of ratings received from previous projects (*#OfRatings*); (3) Project characteristics such as the length of the project description (*DescriptionLength*), the maximum budget specified by the employer (*MaxBid*), posted project duration selected at project creation (*ProjectDuration*), and project-category indicators (*ProjectType*); and (4) Bid-level characteristics such as *BidAmount* and *SameCountry*, an indicator variable equal to one when the employer and worker are located in the same country. Table 1 reports variable definitions and summary statistics, while E-Companion A presents the correlation matrix.

Table 1. Variable Definitions, Measurements, and Summary Statistics

Variable	Description	N	Mean	S.D.	Min	Max
	Project Outcomes and Post-Hiring Outcomes					
AfterChange	A dummy variable for whether the project was posted after the regime-change date	1,926	0.750	0.433	0	1
#OfBidders	Logged number of bidders for a project	1,926	1.872	0.860	0.693	4.860
AvgBidAmount	Logged average bid amount of a project	1,926	4.951	1.651	1.099	17.332
ProjectSuccess	A dummy variable that equals one if the employer chose a worker for a posted project	1,926	0.416	0.493	0	1
TimeToAccept	Logged number of hours the employer took to decide to accept a bid	802	3.317	1.843	0.036	8.929
WorkerRating	A rating given by the employer to the worker after the project is completed (left truncated at 7)	802	9.729	0.714	7	10
Rehire	A dummy for whether the employer hired the same worker again in future projects	802	0.269	0.444	0	1
	Employer Information					
EmployerExperience	Logged number of projects the focal employer has completed at the time of posting the current project	1,926	0.342	0.738	0	3.584
Region	A set of dummy variables to show the regions where the employer comes from (detailed information in E-Companion A)					
	Project Information					
DescriptionLength	Logged length of a project description (i.e., total number of words).	1,926	3.897	1.132	0	6.094
MaxBid	Logged max bid an employer would like to accept	1,926	1.754	2.371	0	11.51

ProjectDuration	Logged posted project duration, measured as the number of days selected by the employer at project creation.	1,926	2.639	0.828	0.693	7.551
ProjectType	A group of dummy variables for the type of projects (detailed information in E-Companion A)					
	Worker Information					
#OfRatings	Logged number of ratings a worker has at the time of the current bid	16,581	0.219	0.578	0	3.761
NoRating	An indicator that equals 1 if the worker is a new worker at the time of the current bid	16,581	0.743	0.437	0	1
WorkerExperience	Logged number of projects the worker has completed at the time of the current bid	16,581	0.329	0.656	0	3.892
	Bid Information					
BidAmount	Logged bid amount for a project	16,581	4.887	1.592	1.099	18.42
BidTime	Logged elapsed time in hours between project posting and bid submission	16,581	3.148	1.867	0	10.73
SameCountry	A dummy variable for whether the worker and employer come from the same country	16,581	0.201	0.401	0	1

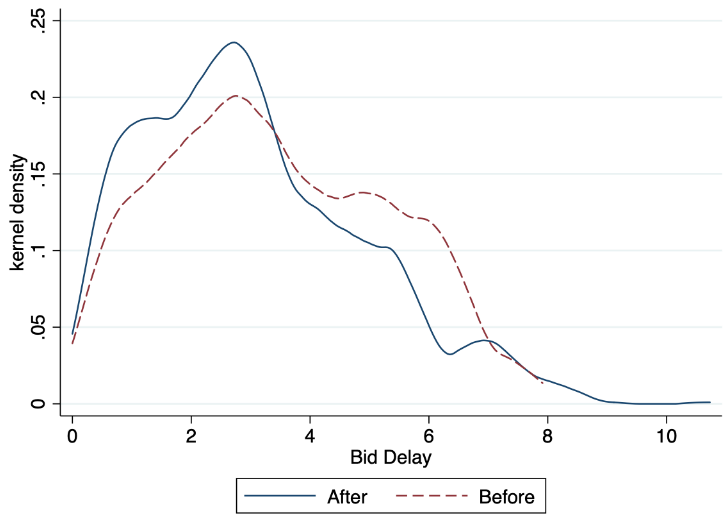
5. Bid Visibility and Bid Timing

The theoretical framework proposes that bid visibility affects market-clearing efficiency and post-award match quality by changing when workers submit bids. We therefore begin the empirical analysis by examining whether the platform's shift from open to sealed bidding reduces strategic delay and induces earlier bid submission. As argued in Section 3, open bidding creates an informational benefit to waiting because workers can observe competing bids before submitting their own offers. Sealed bidding reduces this benefit while leaving the risk of early employer selection unchanged. This analysis is important because bid timing represents the behavioral channel through which the visibility change can alter the flow of offers available to employers.

We measure *BidTime* as the elapsed time in hours between project posting and bid submission. Because the distribution of *BidTime* is highly skewed, we use its logarithmic transformation in the empirical analyses. The analysis is conducted at the bid level and is conditional on a worker submitting a bid. Thus, the estimates capture how the visibility regime affects the timing of observed bids, while changes in whether workers participate are examined separately in later analyses.

Figure 2 provides model-free evidence on this timing response. The distribution of logged *BidTime* shifts downward after the regime change, indicating that bids arrive earlier when competing bids are no longer visible. We further apply a non-parametric two-sample Kolmogorov-Smirnov test to compare the pre- and post-change distributions. The test rejects equality of the distributions, confirming that the distribution of logged *BidTime* differs significantly before and after the visibility change (p-value < 0.001).

Figure 2. Model-Free Evidence (Distribution of *BidTime)*



To examine this relationship more formally, we estimate the following bid-level regression model: *i* denotes the project, *j* denotes the employer, *k* denotes the worker, and *t* denotes the time (the day the worker places the bid).

$$BidTime_{ijkt} = \beta_0 AfterChange_t + \beta X_{ijt} + \delta Z_{ijkt} + \alpha_k + \gamma_t + \varepsilon_{it},$$

where:

$$\beta X_{ijt} = \beta_1 EmployerExperience_{jt} + \beta_2 DescriptionLength_{it} + \beta_3 ProjectType_{it}$$
$$\delta Z_{ijkt} = \delta_1 WorkerExperience_{kt} + \delta_2 BidAmount_{ijkt} + \delta_3 SameCountry_{ijkt} \qquad (1)$$

The vector X_{ijt} includes project and employer characteristics, and the vector Z_{ijkt} includes worker- and bid-level characteristics. We include worker fixed effects, α_k, to control for time-invariant differences across workers in bidding speed, attentiveness, and platform engagement. We also include weekday dummies (i.e., Tuesday to Sunday, denoted by γ_t) to control the day-of-week effects, and project-type dummies to account for systematic temporal and project-category differences in bidding behavior. The coefficient of interest is β_0. A negative estimate indicates that bids are submitted sooner after project posting under sealed bidding than under open bidding. Because the dependent variable is logged, the coefficient can be interpreted as the percentage change in bid submission time associated with the visibility change. Standard errors are clustered at the worker level to account for correlation in bidding behavior across multiple bids submitted by the same worker.

Table 2 reports the results. In column 1, the coefficient on *AfterChange* is negative and statistically significant, indicating that workers submit bids substantially earlier after the platform transitions to sealed bidding. The estimate implies that workers spend approximately 53.0% less time before submitting a bid following the regime change. This result is consistent with our argument that sealed bidding reduces the informational benefit of waiting while preserving the risk that employers may settle down project contracts before delayed bids arrive. Therefore, Hypothesis 1 is supported.

Columns 2 and 3 examine heterogeneity in the bid timing response. Column 2 shows that workers without prior ratings submit bids later on average, consistent with the idea that less established workers face greater uncertainty when evaluating project opportunities. The negative interaction between *AfterChange* and *NoRating* indicates that the timing effect of sealed bidding is stronger for workers without ratings. Column 3 shows that the interaction between *AfterChange* and *#OfRatings* is positive and significant, indicating that the timing effect becomes weaker as workers accumulate more ratings. Together, these results suggest that less established workers are more sensitive to changes in bid visibility. In contrast, experienced workers can rely more on accumulated platform knowledge, reputation, and prior bidding experience when deciding when to participate.

Table 2. Effects of Visibility Change on Bid Time

Dep. Variable	Log Bid Time		
AfterChange	-0.754***	-0.334**	-0.812***
	(0.075)	(0.129)	(0.076)
NoRating		0.664***	
		(0.137)	
AfterChange×NoRating		-0.553***	
		(0.144)	
#OfRatings			0.045
			(0.173)
AfterChange×#OfRatings			0.335**
			(0.107)
EmployerExperience	-0.106***	-0.104***	-0.103***
	(0.019)	(0.019)	(0.019)
WorkerExperience	0.003	0.054	-0.299*
	(0.064)	(0.081)	(0.119)
DescriptionLength	-0.004	-0.003	-0.004
	(0.013)	(0.013)	(0.013)
BidAmount	0.129***	0.128***	0.127***
	(0.012)	(0.012)	(0.012)
SameCountry	-0.015	-0.017	-0.013
	(0.041)	(0.041)	(0.041)
Observations	16,581	16,581	16,581
Number of Workers	3,421	3,421	3,421
Adjusted R-squared	0.058	0.063	0.061
Worker Fixed Effects	YES	YES	YES
Weekday Fixed Effects	YES	YES	YES
Project Type Dummies	YES	YES	YES

Note: Robust standard errors are in parentheses. *** p<0.001, ** p<0.01, * p<0.05

All in all, the model-free distributional evidence and regression results indicate that removing bid visibility changes the timing of worker participation. This finding provides empirical support for the first step in our theoretical framework: bid visibility affects not only whether and how workers compete, but also when their offers become available to employers.

6. Market-Clearing Efficiency

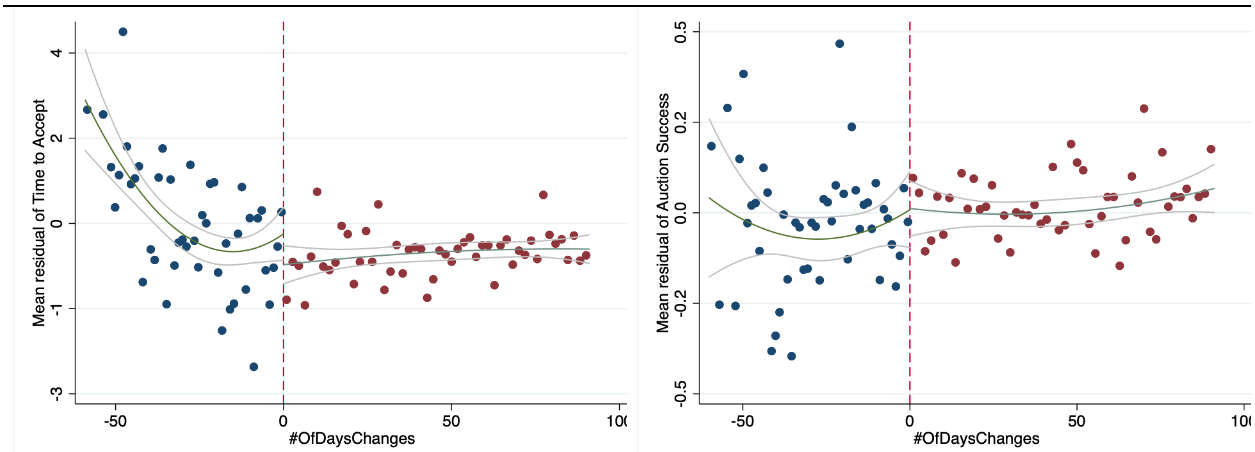
Building on the theoretical framework, we next examine whether changes in bid visibility are associated with improvements in market-clearing efficiency. Section 5 shows that workers submit bids earlier after the platform shifts from open to sealed bidding. If earlier bid arrival increases employers' access to viable candidates, sealed bidding should help employers complete the procurement process more quickly and increase the likelihood that posted projects result in contracts. We therefore examine two outcomes: the time until an employer accepts a bid, measured by *TimeToAccept*, and whether the project results in a contract award, measured by *ProjectSuccess*.

These two outcomes capture distinct dimensions of market-clearing efficiency. *TimeToAccept* captures the speed of market clearing among projects that ultimately result in contracts, whereas *ProjectSuccess* captures the probability that a posted project clears at all. Examining both outcomes is important because removal of visibility could accelerate successful projects without increasing the overall contract formation rate or increase contract formation without substantially reducing time to award.

6.1 Main Analyses and Results

Figure 3. Market-Clearing Efficiency

(a). Time to Accept (b). Project Success



Before turning to regressions, we provide model-free evidence. Figure 3(a) and Figure 3(b) plot residualized market-clearing outcomes before and after the regime change, after controlling for project and employer characteristics. Figure 3(a) suggests a visible decline in the time required to select a worker following the transition to sealed bidding. Figure 3(b) provides weaker visual evidence regarding contracting success, although the fitted values are generally higher after the regime change.

To estimate these effects formally, we use the following project-level model:

$$Outcome_{ijt} = \beta_0 AfterChange_t + \beta X_{ijt} + \alpha_j + \gamma_t + \varepsilon_{it},$$

where:

$$\beta X_{ijt} = \beta_1 EmployerExperience_{jt} + \beta_2 DescriptionLength_{it} + \beta_3 MaxBid_{it} + \beta_4 ProjectDuration_{it} + \beta_5 Region_{jt} + \beta_6 ProjectType_{it} \qquad (2)$$

In regression equation (2), *Outcome* is the dependent variable, representing *TimeToAccept* and *ProjectSuccess*, respectively. The primary independent variable is *AfterChange*, which equals 1 if the project posting day *t* is after the visibility change. Since the regression model is at the project level, we only control for project and employer-specific characteristics measured at time *t*, which is denoted by the covariate set X_{ijt}. Additionally, we incorporate weekday dummies to control for day-of-week effects. For *TimeToAccept*, we take a logarithmic transformation to avoid skewed distributions. We estimate employer random effects models rather than employer fixed effects models, due to a limited number of employers posting at least two projects, which precludes sufficient within-employer variation. We use a random effects linear probability model (LPM) for the dependent variable *ProjectSuccess*. Standard errors are clustered at the employer level.

Table 3. Effects of Visibility Change on Market Efficiency

Dep. Variable	*Time to Accept*	*Project Success*
AfterChange	-0.493**	0.061*
	(0.163)	(0.028)
EmployerExperience	-0.354***	0.068
	(0.080)	(0.042)
MaxBid	-0.049+	0.011*
	(0.025)	(0.005)
DescriptionLength	0.145**	0.017
	(0.050)	(0.012)
ProjectDuration	1.115***	-0.124***
	(0.095)	(0.015)
Observations	802	1,926
Number of Employers	423	967
Adjusted R^2	0.306	0.091
Employer Region Dummies	YES	YES
Weekday Dummies	YES	YES
Project Type Dummies	YES	YES

Note: Robust standard errors are in parentheses. *** p<0.001, ** p<0.01, * p<0.05

Table 3 reports the main results. Column 1 shows that the coefficient of *AfterChange* is negative and statistically significant when *TimeToAccept* is the dependent variable. Because *TimeToAccept* is measured in logarithmic form, the coefficient implies that projects posted after the removal of bid visibility experience approximately 38.9% shorter times to contract award. This result supports Hypothesis 2 and suggests that employers make hiring decisions substantially more quickly after competing bids become hidden.

Column 2 shows that the coefficient on *AfterChange* is positive and statistically significant when *ProjectSuccess* is the dependent variable. The estimate implies an increase of approximately 6.1% points in the probability that a posted project results in a contract. This result supports Hypothesis 3 and suggests that sealed bidding increases the likelihood of market clearing, not merely the speed of already-successful matches.

All these results indicate that the visibility change is associated with improvements in both dimensions of market-clearing efficiency: employers award contracts faster, and a larger share of posted projects result in contract formation.

6.2 Identification, Robustness, and Heterogeneity Analyses

The previous results suggest that removing bid visibility improves market-clearing efficiency by reducing the time to contract award and increasing the likelihood of project success. In this section, we conduct additional analyses to assess the robustness and boundary conditions of these findings. Because the policy change was implemented platform-wide, our setting does not include an untreated control group. The main identification concern is whether the estimated effects reflect the visibility change itself rather than contemporaneous changes in project, employer, or worker composition, unobserved temporal shocks, or heterogeneous responses across market participants. We address these concerns through a series of identification, robustness, and heterogeneity analyses, summarized in Table 4. Detailed results are reported in the E-Companion B.

First, we examine whether observable market composition changed discontinuously around the regime-change date. If the post-change projects, employers, or workers differed sharply from those in the pre-change period, the estimated effects could reflect compositional shifts rather than the removal of bid visibility. We therefore conduct discontinuity-style composition tests using observable project, employer, and worker characteristics as dependent variables. Specifically, we examine employer experience, employer contract value, project budget, project description length, average worker experience, and average worker contract value around the regime-change date. The graphical evidence and regression results, reported in E-Companion B-1, show no systematic discontinuities in these observable characteristics at the cutoff. These results suggest that the policy change was not accompanied by abrupt changes in the observable composition of projects, employers, or participating workers.

Second, we conduct placebo tests using fake regime-change dates. These tests address the possibility that the main results are driven by general time trends, seasonality, or other unobserved shocks rather than the actual visibility change. We re-estimate the main specifications after assigning placebo cutoff dates around the actual regime-change date.[2] If spurious temporal patterns drove our findings, similar effects should appear around these fake cutoffs. The placebo results, reported in E-Companion B-2, show no statistically significant effects around the placebo dates for the key market-clearing outcomes. This pattern supports the interpretation that the observed changes in *TimeToAccept* and *ProjectSuccess* are linked to the actual visibility change rather than to unrelated temporal shifts.

Third, we further assess whether the results are sensitive to sample balancing and estimator choice. To address potential imbalance between pre- and post-change projects, we estimate matching-based specifications using propensity score matching and nearest-neighbor matching. These analyses compare projects that are similar in observable characteristics but differ in whether they were posted before or after the visibility change. The results, reported in E-Companion B-3, remain consistent with the baseline findings: sealed bidding is associated with faster contract awards and a higher probability of project success.

Fourth, we estimate alternative model specifications for the dependent variables. For *TimeToAccept*, we use a Cox proportional hazards model, which treats contract award as a time-to-event outcome. For *ProjectSuccess*, we estimate random-effects Logit and Probit models. The results, reported in E-Companion B-4, are qualitatively unchanged. These analyses indicate that the main findings are not artifacts of the linear model specification.

Finally, we examine whether the effects of the visibility change vary across market participants and project contexts. These analyses are theoretically relevant because the value of bid visibility should depend on participants' experience and the nature of the procurement task. More experienced employers may process incoming offers more efficiently and may be less dependent on bid visibility when evaluating candidates. Similarly, more experienced workers may rely less on competitors' visible bids because they have accumulated knowledge about project fit, pricing, and employer preferences. The heterogeneity analyses, reported in E-Companion B-5, show that the market-clearing effects vary with employer experience and average worker experience. The reduction in *TimeToAccept* and the increase in *ProjectSuccess* are smaller for more experienced employers. Average worker experience also weakens the reduction in *TimeToAccept,* although it does not significantly moderate the effect on *ProjectSuccess*. These results indicate that the consequences of removing bid visibility are not uniform across market conditions; rather, the benefits of sealed bidding appear stronger when participants are more sensitive to

[2] Specifically, we apply the placebo cutoff dates as 7 or 14 days before or after the actual regime-change date. For the pre-change placebo tests (-7 or -14), we only use observations before the actual regime-change date; for the post-change placebo tests (+7 or +14), we only use observations after the actual regime-change date.

Table 4. Summary of Identification, Robustness, and Heterogeneity Analyses

#	Analysis	Dependent Variables	Concern Addressed	Method	Main Findings	E-Companion
1	Composition tests	*Employer Experience, Employer Contract Value, Project Description Length, Project Budget, Average Worker Experience, Average Worker Contract Value*	Changes in market composition around the policy change	Discontinuity tests around the regime-change date	No systematic discontinuities in observable employer, worker, or project characteristics	B-1
2	Placebo tests	*TimeToAccept, ProjectSuccess*	Unobserved time-varying shocks or seasonal trends	Re-estimation using placebo regime-change dates	No significant effects around placebo cutoffs	B-2
3	Matching estimators	*TimeToAccept, ProjectSuccess*	Sample composition differences between pre- and post-change projects	Nearest-neighbor matching / propensity-score matching	Results remain consistent after matching	B-3
4	Alternative estimators	*TimeToAccept*	Functional-form sensitivity	Cox proportional hazards model	Results remain qualitatively unchanged	B-4
		ProjectSuccess		Random-effects Logit and Probit models		
5	Employer experience	*TimeToAccept, ProjectSuccess*	Heterogeneous employer responses	Interaction models	Effects vary by employer experience	B-5(a)
6	Worker experience	*TimeToAccept, ProjectSuccess*	Heterogeneous worker responses	Interaction models	Effects vary by average worker experience	B-5(b)

the information environment and timing incentives.

Taken together, these additional analyses strengthen the interpretation of the main results. The composition, placebo, matching, and alternative-estimator analyses reduce concerns that observable composition changes, spurious time trends, sample imbalance, or modeling choices drive the market-clearing findings. The heterogeneity analyses further clarify the conditions under which the visibility change matters most. Overall, the evidence is consistent with our main argument that reduced bid visibility is associated with faster contract awards and a higher probability of project success, with stronger effects in settings where participants are more sensitive to the information environment.

Having examined the main market-clearing outcomes and supporting timing evidence, we next consider whether changes in competitive structure or post-award match quality accompany these efficiency gains.

7. Competitive Tradeoffs and Match Quality

The results in Section 6 show that sealed bidding is associated with faster contract awards and a higher probability of contract formation. However, improved market-clearing efficiency does not necessarily mean that the platform produces better matches. Faster contracting may be achieved through changes in the competitive landscape of the bidding process, such as the number of participating workers, average bid amounts, and opportunities for bid revision. These changes are important to examine before evaluating post-award match quality because they clarify whether sealed bidding accelerates market clearing by weakening competition, by restructuring competition among participating workers, or by improving the timing with which employers receive actionable offers. We therefore first examine participation and average bid amounts as competitive outcomes and then assess whether contracts formed under sealed bidding generate comparable or better post-award match quality.

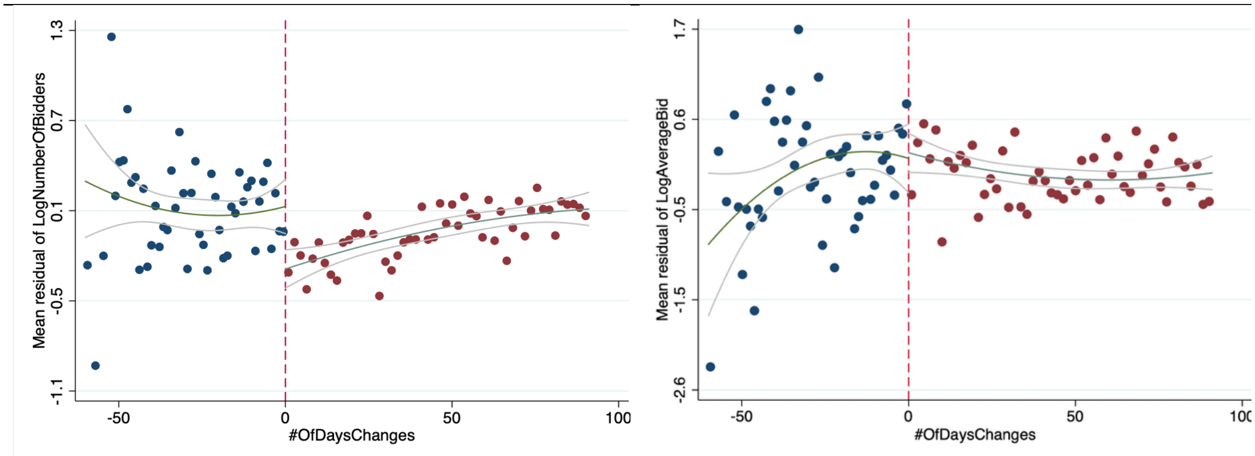
7.1 Participation and Competition

We first examine two supplementary outcomes for competition: worker participation and average bid amount. Participation is measured by the number of unique workers bidding on a project (*#OfBidders*). Average bid amount is measured by the average wage proposed by participating workers (*AvgBidAmount*). These outcomes are not formal hypotheses in our framework, but they help characterize the tradeoffs associated with the visibility change. This distinction is important because platform design may affect not only the number of participants but also the composition of participants attracted to a market opportunity (Chen et al. 2021). If sealed bidding reduces chances for strategic monitoring and bid revision, fewer workers may participate, while participating workers may submit more competitive offers (with lower average bid amount).

Before turning to the regression analysis, we provide model-free evidence. Figure 4(a) and Figure 4(b) plot residualized log-transformed *#OfBidders* and *AvgBidAmount* before and after the regime change, after controlling for project and employer characteristics.[3] Figure 4(a) shows a noticeable decline in participation following the transition to sealed bidding. Figure 4(b) suggests that average bid amounts become lower after the regime change, indicating more aggressive bidding among participating workers.

Figure 4. Participation and Average Bid Amount

(a). Logged *#OfBidders* (b). Logged *AvgBidAmount*



We further formally examine these relationships using the same project-level regression model used in the market-clearing analyses, and the results are reported in Table 5. Column 1 shows that the coefficient on *AfterChange* is negative and statistically significant, indicating that the number of unique bidders per project declines by approximately 10.3% after the regime change. This result suggests that sealed bidding is associated with a smaller realized pool of participants. Column 2 shows that average bid amounts are significantly lower after the regime change. Because the dependent variable is measured in logarithmic form, the coefficient implies that average bid amount is approximately 19.0% lower after the transition to sealed bidding. This finding suggests that workers who participate submit more aggressive wage offers when they cannot observe competing bids.

These results indicate that sealed bidding changes the patterns of participation and competition. In particular, fewer observed bidders do not necessarily imply weaker price competition. To better understand the timing and composition of participation, we examine whether sealed bidding changes the early arrival of bidders at the project level, focusing on both all bidders and experienced bidders. These analyses, reported in E-Companion C-1, show that sealed bidding is associated with a higher probability

[3] Because both *#OfBidders* and *AvgBidAmount* are highly skewed, we apply logarithmic transformations in the analyses.

that a project receives at least one bid within the first 24 hours, both from any bidder and from experienced bidders. Sealed bidding is also associated with a larger share of bidders and experienced bidders arriving within the first 24 hours. Notably, sealed bidding does not reduce the number of bidders arriving within the first 24 hours; it even increases the number of experienced bidders arriving within this early window. These results provide additional evidence that, although sealed bidding reduces overall worker participation, it increases the likelihood that employers receive timely and actionable offers early in the procurement process.

Table 5. Effects of Visibility Change on Competition

Dep. Variable	*#OfBidders*	*AvgBidAmount*
AfterChange	-0.109*	-0.211*
	(0.047)	(0.098)
EmployerExperience	-0.144***	-0.048
	(0.039)	(0.079)
MaxBid	-0.013	-0.056**
	(0.009)	(0.018)
DescriptionLength	0.142***	0.113**
	(0.018)	(0.041)
ProjectDuration	0.130***	0.363***
	(0.028)	(0.056)
Observations	1,926	1,926
Adjusted R^2	0.145	0.112
Number of Employers	967	967
Employer Region Dummies	YES	YES
Weekday Dummies	YES	YES
Project Type Dummies	YES	YES

Note: Robust standard errors are in parentheses. *** p<0.001, ** p<0.01, * p<0.05

We further examine repeated bidding activity as supplementary evidence on how sealed bidding changes competitive behavior. Under open bidding, workers can observe competitors' bids after entering the auction, which creates opportunities for strategic monitoring and bid revision. Workers may therefore submit an initial bid, continue observing rivals, and subsequently revise their offers as new competitive information becomes available. Sealed bidding reduces this informational value of monitoring because competing bids are no longer visible, making repeated bid adjustment less useful. Consistent with this mechanism, E-Companion C-2 shows that both the total number of bids submitted to a project *(#OfBids*) and the number of bids submitted per bidder *(#BidsPerBidder*) decline significantly after the regime change. These findings suggest that sealed bidding reduces repeated bidding and limits opportunities for strategic bid adjustment.

These competitive tradeoffs help clarify the market-clearing results reported in Section 6. Sealed bidding appears to reduce the volume of observed worker participation, but it also accelerates bid arrival and is associated with lower average bid amounts. These changes may increase the likelihood that employers encounter acceptable offers earlier in the procurement process, thereby facilitating faster contract awards and a higher probability of contract formation.

7.2 Match Quality

The results above show that sealed bidding is associated with market-clearing efficiency and restructures worker participation. An important remaining question is whether these operational gains come at the expense of post-award match quality. On the one hand, faster contracting could reduce match quality if employers make decisions from a smaller candidate pool or have less opportunity to evaluate workers carefully. On the other hand, since sealed bidding encourages earlier arrival of viable offers, employers may be able to screen candidates sooner and select suitable workers without sacrificing match quality. Therefore, the impact of removal of bid visibility on match quality is an empirical question.

To examine this question, we focus on employer satisfaction with completed projects. Specifically, we use the rating assigned by the employer to the winning worker upon project completion (*WorkerRating*). Employer numerical ratings (from 0 to 10 in our context) are commonly used as post-transaction performance measures in online platforms. Because ratings below 7 are rare, we left-censor *WorkerRating* at 7, resulting in a scale from 7 to 10.[4] Let $EmployerSatisfaction_{ijt}$ denote the post-project outcome of project *i* posted by employer *j* at time *t*. We propose the following regression model:

$$EmployerSatisfaction_{ijt} = \beta_0 AfterChange_t + \beta X_{ijt} + \varepsilon_{it},$$

where:

$$\beta X_{ijt} = \beta_1 DescriptionLength_{it} + \beta_2 EmployerExperience_{jt} + \beta_3 WorkerExperience_{jt} + \beta_4 SameCountry_{it} + \beta_5 WinningBid_{it} + \beta_6 ProjectType_{it} \quad (3)$$

This analysis focuses exclusively on contracted projects in which employers have selected a winning worker. We employ a linear regression. We estimate three model specifications. The first specification includes only the main independent variable (*AfterChange*) to show the baseline association with the visibility change. The second specification adds project characteristics such as *DescriptionLength*, *WinningBid*, and *ProjectType* dummies. Finally, the third specification further includes employer and worker experience.

Table 6 reports the results. The coefficient on *AfterChange* is positive and statistically significant across the specifications, indicating that employers report higher satisfaction with projects completed

[4] In our data, only 2.99% of observations will be affected by truncation, since the worker gets a rating below 7 in 24 out of 802 completed projects.

after the transition to sealed bidding. The estimate remains positive after controlling for project, employer, and worker characteristics. This finding suggests that the improvement in market-clearing efficiency does not come at the expense of observed post-award match quality. Instead, the evidence is consistent with Hypothesis 4: contracts formed under sealed bidding are associated with better employer-rated match quality than contracts formed under open bidding.

Table 6. Effects of Visibility Change on Employer Satisfaction

Dep. Variable		Worker Rating	
AfterChange	0.249***	0.245***	0.205**
	(0.0858)	(0.0852)	(0.0866)
DescriptionLength		0.00456	0.0116
		(0.0250)	(0.0246)
EmployerExperience			0.0543**
			(0.0227)
WorkerExperience			0.0526**
			(0.0239)
SameCountry			0.0252
			(0.0605)
WinningBid		-0.0121	-0.0160
		(0.0209)	(0.0215)
Observations	802	802	802
Adjusted R^2	0.019	0.025	0.029
Project-type Dummies	No	YES	YES

Note: Robust standard errors are in parentheses. *** p<0.001, ** p<0.01, * p<0.05

We further assess the robustness of the *WorkerRating* results using ordered Logit and ordered Probit models, and the results remain qualitatively similar. We also examine *Rehire* as an additional measure of match quality, since employers who are satisfied with the workers' performance will be more likely to hire the same workers in future projects. The estimated effects on *Rehire* are positive but sensitive to model specification and become statistically insignificant after employer and worker characteristics are included. We therefore interpret the rehiring result as suggestive rather than conclusive evidence. These additional analyses are reported in E-Companions C-3 and C-4. In sum, the match-quality results suggest that sealed bidding is associated with higher employer-rated match quality among completed projects.

Overall, the results in this section suggest that sealed bidding changes the competitive structure of the procurement process without undermining observed post-award match quality. Although sealed bidding is associated with fewer overall bidders, participating workers submit lower average bid amounts, and the additional early-arrival analyses show that employers are more likely to receive timely offers, especially from experienced workers. Sealed bidding also reduces repeated bidding activity, consistent with the idea that hiding competing bids limits opportunities for strategic monitoring and bid revision. Thus, the decline in overall participation should not be interpreted mechanically as weaker competition. Rather, the evidence suggests that sealed bidding restructures competition by shifting worker behavior away from delayed monitoring and repeated adjustment and toward earlier, more actionable, and more competitive offers. Consistent with this interpretation, the match-quality results show higher employer-rated match quality among completed projects. Taken together, these findings support the timing mechanism proposed in our theoretical framework: reducing bid visibility lowers the informational value of waiting, induces earlier bid submission, improves market-clearing speed and contract formation, and does so without reducing observed post-award match quality.

8. Discussion

8.1 Summary of Findings

This study examines how a platform-wide change in bid visibility reshaped bid timing, market-clearing efficiency, competitive structure, and post-award match quality in an online service procurement marketplace. After competing bids became hidden, workers submitted bids earlier, employers awarded contracts sooner, and a larger share of posted projects resulted in contract formation. At the same time, realized worker participation declined, and average submitted wages fell. Regarding post-award match quality, both employer ratings and rehiring rates are higher after the removal of bid visibility, although the positive rehiring effect was not robust to the full set of employer and worker controls. Overall, the results suggest that the visibility change primarily altered the timing of participation and the efficiency with which employers and workers reached a match.

The evidence is consistent with a strategic-timing mechanism. When competing bids are visible, workers can acquire information by delaying participation and observing rivals before submitting an offer. However, because employers may award projects before a posted deadline, waiting is also associated with the risk of losing the opportunity to compete. Sealed bidding reduces the informational benefit of delay while preserving the opportunity cost of waiting. This change alters the value of waiting and encourages earlier bid submission. More broadly, the findings suggest that information visibility affects procurement outcomes not only through learning, participation, and price competition, but also through its influence on the strategic bid-timing decisions of workers.

8.2 Theoretical Contributions

This study mainly contributes to the literature in three ways. First, we extend research on information visibility and disclosure in procurement markets by identifying bid timing as an important but previously underexamined consequence of information policies. Existing research has primarily focused on how information revelation affects bidder learning, participation, prices, surplus, and allocation outcomes (Arora et al. 2007; Greenwald et al. 2010; Kannan 2012; Haruvy and Katok 2013; Lu et al. 2019). In these studies, transparency is typically evaluated according to the information it provides and the effects on competition and allocation. Our findings suggest an additional mechanism. By changing the informational value of waiting, visibility policies influence not only what market participants know but also when they choose to act. In environments characterized by asynchronous arrival and employer-controlled stopping, information disclosure therefore affects operational outcomes through its impact on participation timing. This perspective broadens existing theories of information visibility by highlighting timing as an important behavioral channel linking disclosure policies to procurement outcomes.

Second, we bridge information disclosure and bid-timing literatures by showing that information visibility shapes incentives to delay participation. Prior research has shown that bid timing is influenced by auction-ending rules, opportunities for learning, and the costs of waiting (Roth and Ockenfels 2002; Ockenfels and Roth 2006; Ely and Hossain 2009; Carare and Rothkopf 2005; Katok and Kwasnica 2008). However, this literature has paid relatively little attention to how information environments shape the timing of bidder actions. By integrating the information disclosure and bid-timing literatures, we show that visibility itself can create incentives for strategic delay because workers can learn from competitors before submitting offers. The evidence indicates that removing bid visibility reduces the informational benefit of waiting and leads workers to submit bids earlier. More broadly, our findings suggest that information design and timing behavior should be studied jointly rather than as separate dimensions of market design.

Third, we contribute to the literature on online service procurement and buyer-determined service markets by shifting attention from static allocation outcomes to dynamic market-clearing efficiency. Prior studies of online labor markets and buyer-determined procurement have focused primarily on participation, bid amounts, buyer surplus, matching decisions, and allocation efficiency (Che 1993; Asker and Cantillon 2008; Hong et al. 2016; Liang et al. 2022). Implicitly, these studies often evaluate procurement performance based on the set of offers that ultimately arrive. Our findings highlight the importance of a complementary operational perspective. In markets where workers arrive asynchronously, and employers may award projects before a formal deadline, procurement performance depends not only on the quality of observed bids but also on the speed with which acceptable offers emerge. By showing that visibility policies are associated with changes in time-to-contract, contract formation, and observed post-award match quality, we extend existing theories of online service procurement from static allocation efficiency toward dynamic matching and market-clearing efficiency.

All contributions suggest a broader principle for digital procurement markets: information policies influence market performance not only by shaping information flows and competitive behavior, but also by altering the timing of participation. Understanding this timing channel is essential for explaining how procurement mechanisms affect operational outcomes in online service markets.

8.3 Managerial Implications

Our findings suggest that platform management should view information disclosure and process timing as interdependent design choices rather than independent features of procurement systems. Traditional discussions of transparency focus on improving information quality, bidder learning, and competitive outcomes. However, our results indicate that transparency can also influence when participants act. In environments where suppliers arrive asynchronously and buyers may terminate procurement before a formal deadline, information policies affect not only information acquisition but also the speed at which actionable offers emerge. Management should therefore evaluate disclosure policies together with decision deadlines, stopping rules, and participation dynamics.

The results are particularly relevant for organizations that procure customized services through digital platforms. Examples include online labor markets, professional-service procurement systems, crowdsourcing contests, consulting engagements, and corporate supplier-sourcing platforms. In these settings, managers often face a tradeoff between encouraging information sharing and accelerating decision making. Our findings suggest that reducing bid visibility may improve market-clearing efficiency when suppliers can strategically delay participation and buyers benefit from receiving acceptable offers sooner. Conversely, greater transparency may remain desirable when price discovery, bidder learning, or common-value uncertainty are primary concerns. The optimal level of transparency therefore depends on both informational and operational considerations. Thus, managers should not interpret sealed bidding as universally superior; rather, reduced visibility is most attractive when delayed participation is a major operational bottleneck and early availability of acceptable offers is more valuable than incremental bidder learning.

More broadly, our findings suggest that procurement performance should not be evaluated solely using traditional metrics such as participation levels, prices, or bidder counts. A common assumption is that a larger bidder pool necessarily reflects a healthier marketplace. Our results challenge this view by showing that fewer observed participants can coexist with faster contracting, higher contract-formation rates, lower proposed wages, and higher employer ratings. Managers should therefore adopt a multidimensional perspective when evaluating procurement mechanisms. This implication is consistent with related operations management research on crowdsourced operations, which shows that platform design and communication can shape participation, task quality, and worker satisfaction (Ta et al. 2021). In addition to participation and prices, organizations should monitor operational metrics such as time to first viable offer, time to contract award, project contracting rates, employer satisfaction, and future repeat contracting. These measures provide a more comprehensive assessment of marketplace performance and procurement effectiveness.

Although our analysis focuses on online labor markets, the underlying mechanism may apply to a broader class of digital matching environments. Many contemporary platforms, including supplier marketplaces, crowdsourcing platforms, internal talent marketplaces, and B2B procurement systems, rely on participants arriving over time while decision makers retain discretion regarding when to stop searching and commit to a match. In these settings, market designers must balance the informational benefits of transparency against the operational benefits of timely participation. Our findings suggest that managing this tradeoff is important for organizations seeking to improve both match quality and market efficiency.

8.4 Limitations and Future Research

This study has several limitations that suggest directions for future research. First, the analysis is based on a platform-wide policy change, which limits causal identification because no contemporaneous control group exists. Although our results are robust across multiple specifications and validity checks, future research could exploit randomized disclosure policies or staggered rollouts to provide stronger causal evidence. Second, our data capture observed bidding behavior but not earlier stages of the participation process, such as project search, project viewing, or workers' decisions not to bid. Future work could examine how information visibility influences the broader sequence of search, attention, participation, and bidding decisions. Third, because the evidence comes from a single online labor marketplace, additional studies across different procurement settings and disclosure designs are needed to assess the generalizability of the proposed timing mechanism.

Finally, our analysis focuses on short-run responses around the visibility change. Workers and employers may adapt to sealed bidding over time as they learn the new information environment. Future research could examine longer-run adaptation, including whether the timing effects persist, whether worker participation recovers, and whether platforms can combine selective transparency with mechanisms that preserve timely participation. Such work would further clarify when transparency improves procurement performance and when it creates operational frictions through strategic delay.

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E-Companion for

When Visibility Shapes Timing: Evidence on Bidding Dynamics and Market Efficiency

Contents

E-Companion A: Key Variable Correlation Matrix and Variables

Table A1. Correlations among Key Variables

#	Variable Names	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
1	*AfterChange*	1.000														
2	*#OfBidders*	-0.026	1.000													
3	*ProjectSuccess*	0.062	-0.067	1.000												
4	*TimeToAccept*	-0.159	0.483	n/a	1.000											
5	*WorkerRating*	0.143	0.025	n/a	-0.042	1.000										
6	*EmployerExperience*	0.163	-0.059	0.074	-0.260	0.088	1.000									
7	*DescriptionLength*	0.076	0.104	0.067	0.200	-0.005	-0.035	1.000								
8	*MaxBid*	-0.029	-0.013	0.011	-0.112	-0.062	-0.065	-0.010	1.000							
9	*ProjectDuration*	-0.025	0.301	-0.212	0.404	-0.059	-0.071	-0.057	-0.067	1.000						
10	*#OfRatings*	0.136	-0.107	0.124	-0.163	0.091	0.041	-0.012	0.005	-0.116	1.000					
11	*NoRating*	-0.152	0.125	-0.134	0.185	-0.079	-0.056	0.007	-0.016	0.138	-0.637	1.000				
12	*WorkerExperience*	0.175	-0.126	0.138	-0.182	0.101	0.055	-0.008	0.011	-0.132	0.933	-0.834	1.000			
13	*BidAmount*	-0.037	0.116	-0.233	0.210	-0.026	-0.101	0.095	-0.013	0.213	-0.068	0.052	-0.062	1.000		
14	*BidTime*	-0.096	0.286	-0.287	0.723	0.002	-0.096	0.003	-0.041	0.473	-0.240	0.304	-0.288	0.184	1.000	
15	*BidOrder*	-0.028	0.663	-0.030	0.369	0.033	-0.032	0.040	-0.028	0.270	-0.132	0.176	-0.163	0.072	0.570	1.000
16	*SameCountry*	-0.126	0.006	-0.014	0.029	-0.011	-0.062	-0.001	0.000	0.000	-0.060	0.043	-0.064	0.001	0.016	0.010

Note: *TimeToAccept* and *WorkerRating* are only valid when the project is successfully contracted. Thus, their correlation with *ProjectSuccess* is n/a

Table A2. Distribution of Project Type and Employer Region

Dummy Code	Description	Frequency	Percent
	Project Type*		
1	Website and Software Development	1,869	95.94%
2	Writing and Content	89	4.57%
3	Graphical Design	269	13.81%
4	Data Entry and Management	568	29.16%
	Region		
1	Asian	105	5.39%
2	Europe	398	20.43%
3	North America	945	48.51%
4	South America	17	0.87%
5	Australia/Oceania	34	1.75%
6	Africa	12	0.62%
0	N/A	437	22.43%

*Most projects belong to multiple types; the values are not mutually exclusive

E-Companion B: Identification, Robustness, and Heterogeneity Analyses

B-1: Composition Tests

A key concern in our platform-wide policy-change setting is that projects, employers, or workers may differ before and after the bid visibility change. If the composition of market participants changed abruptly around the regime-change date, the estimated effects in the main analyses could reflect compositional shifts rather than changes in bid visibility. To assess this concern, we conduct composition tests using observable employer, project, and worker characteristics as dependent variables.

Specifically, we estimate discontinuity-style regressions around the regime-change date. The dependent variables include employer experience, employer contract value, project budget, project description length, average worker experience, and average worker contract value. The regressions include the number of days relative to the regime-change date, the post-change indicator, and their interaction. The coefficient on the post-change indicator captures whether there is an abrupt discontinuity in the observable characteristic at the regime-change date, while the time trend and interaction terms capture gradual changes before and after the policy change.

Figure B1 provides graphical evidence, and Table B1 reports the regression results. Overall, the results do not show systematic discontinuities in observable employer, worker, or project characteristics at the cutoff. Although some variables exhibit smooth time trends around the regime change, the absence of broad discontinuous shifts suggests that the main results are unlikely to be driven solely by abrupt changes in market composition. These tests support the use of the platform-wide policy change as a useful setting for examining how bid visibility is associated with bidding behavior and market-clearing outcomes.

Figure B1. Composition Tests

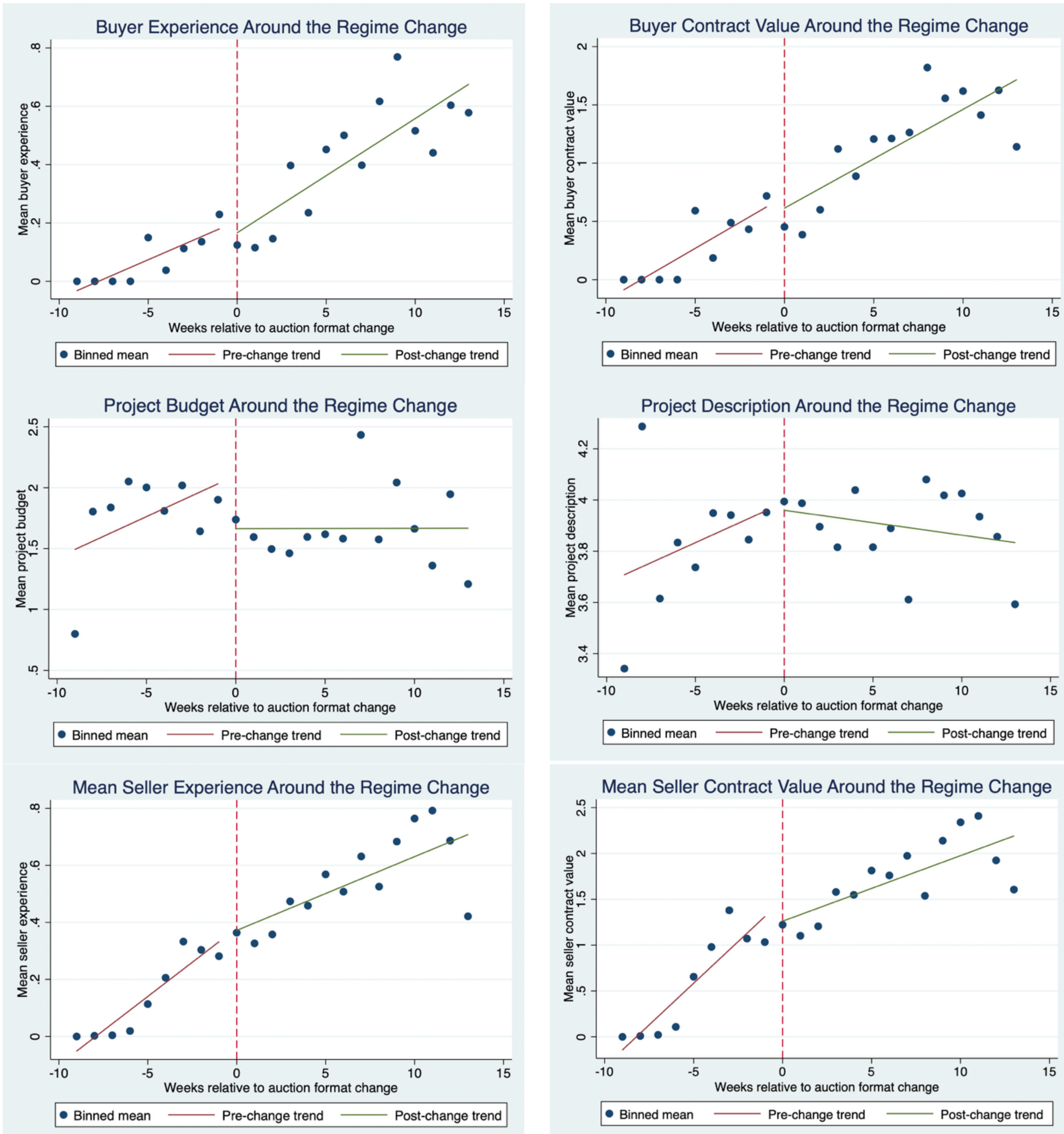

Table B1. Composition Tests Regression

Dep. Variable	Employer Experience	Employer Contract Value	Project Budget	Project Description	Mean Worker Experience	Mean Worker Contract Value
Days to Change	0.005***	0.014***	-0.002	0.005	0.008***	0.027***
	(0.001)	(0.003)	(0.007)	(0.003)	(0.001)	(0.002)
AfterChange	-0.070	-0.188	-0.231	-0.044	-0.043	-0.239*

	(0.049)	(0.141)	(0.247)	(0.112)	(0.039)	(0.119)
Days to Change	0.001	0.000	0.004	-0.005	-0.003**	-0.015***
× *AfterChange*	(0.001)	(0.004)	(0.008)	(0.003)	(0.001)	(0.003)
Constant	0.211***	0.701***	1.842***	3.964***	0.371***	1.413***
	(0.066)	(0.032)	(0.091)	(0.053)	(0.027)	(0.093)
Observations	1,948	1,948	1,948	1,948	1,948	1,948
Adjusted R^2	0.072	0.074	0.001	0.001	0.125	0.135

Note: Robust standard errors are in parentheses. *** p<0.001, ** p<0.01, * p<0.05

B-2: Placebo Tests

We next conduct placebo tests to examine whether general time trends, seasonality, or other temporal shocks unrelated to the actual visibility change could drive the main results. If the estimated effects in the main analyses were driven by spurious temporal patterns, similar effects should appear when the regime-change date is artificially shifted to nearby placebo dates.

We construct placebo indicators using artificial cutoff dates 7 and 14 days before and after the actual regime-change date. For placebo tests before the actual change, we use only observations from the pre-change period. For placebo tests after the actual change, we use only observations from the post-change period. This design avoids mixing observations that were exposed to different bid visibility regimes.

Tables B2a and B2b report the placebo results for *TimeToAccept* and *ProjectSuccess*, respectively. Across the placebo specifications, the placebo indicators are not statistically significant for the key market-clearing outcomes. These results suggest that the main findings are not simply artifacts of local time trends or seasonality around the study period. Instead, the changes in contract-award speed and project success appear more closely associated with the actual platform-wide visibility change.

Table B2a. Effect of Visibility Change on Market Clearing Efficiency: Placebo Test for Time to Accept

Dep. Variable	Time to Accept			
AfterPlacebo (-7 Days)	-0.148			
	(0.327)			
AfterPlacebo (+7 Days)		0.394		
		(0.262)		
AfterPlacebo (-14 Days)			-0.427	
			(0.279)	
AfterPlacebo (+14 Days)				0.290
				(0.194)
EmployerExperience	-0.722*	-0.388***	-0.647*	-0.397***
	(0.289)	(0.103)	(0.283)	(0.106)
MaxBid	-0.086	-0.038	-0.089	-0.040
	(0.063)	(0.029)	(0.063)	(0.030)
DescriptionLength	0.012	0.145**	0.009	0.146**
	(0.131)	(0.052)	(0.130)	(0.052)
ProjectDuration	1.041***	1.102***	1.011***	1.097***
	(0.176)	(0.115)	(0.177)	(0.114)
Observations	169	633	169	633
Number of Employers	116	333	116	333

Employer Region Dummies	YES	YES	YES	YES
Weekday Dummies	YES	YES	YES	YES
Project Type Dummies	YES	YES	YES	YES

Note: Robust standard errors are in parentheses. *** p<0.001, ** p<0.01, * p<0.05

Table B2b. Effect of Visibility Change on Market Clearing Efficiency: Placebo Test for Project Success

Dep. Variable	Project Success			
AfterPlacebo (-7 Days)	-0.073			
	(0.052)			
AfterPlacebo (+7 Days)		0.033		
		(0.050)		
AfterPlacebo (-14 Days)			-0.014	
			(0.050)	
AfterPlacebo (+14 Days)				0.028
				(0.041)
EmployerExperience	0.235**	0.075*	0.219**	0.074*
	(0.082)	(0.037)	(0.077)	(0.036)
MaxBid	0.007	0.013*	0.007	0.013*
	(0.009)	(0.006)	(0.009)	(0.006)
DescriptionLength	0.064**	0.006	0.063**	0.006
	(0.024)	(0.014)	(0.024)	(0.014)
ProjectDuration	-0.102***	-0.128***	-0.102***	-0.128***
	(0.027)	(0.019)	(0.027)	(0.019)
Observations	482	1,444	482	1,444
Number of Employers	318	720	318	720
Employer Region Dummies	YES	YES	YES	YES
Weekday Dummies	YES	YES	YES	YES
Project Type Dummies	YES	YES	YES	YES

Note: Robust standard errors are in parentheses. *** p<0.001, ** p<0.01, * p<0.05

B-3: Matching Estimators

Although the composition tests reduce concerns about abrupt discontinuities, projects posted before and after the visibility change may still differ in observable characteristics. To address this possibility, we estimate matching-based specifications that compare pre- and post-change projects with similar observable characteristics. These analyses provide an additional robustness check for the main market-clearing results.

We use two matching approaches: propensity score matching and nearest-neighbor matching. The matching procedure uses observable employer and project characteristics to construct comparable pre- and post-change samples. The dependent variables are *TimeToAccept* and *ProjectSuccess*, corresponding to the two dimensions of market-clearing efficiency examined in the main paper.

Table B3 reports the matching results. The estimates remain consistent with the baseline findings. Under both propensity score matching and nearest-neighbor matching, sealed bidding is associated with shorter time to contract award and a higher probability of project success. These results indicate that the main findings are not driven by observable differences between projects posted before and after the visibility change.

Table B3. Effect of Visibility Change on Market-Clearing Efficiency: Matching Estimators

Dep. Variable	Market-Clearing Efficiency	
	Time to Accept	Project Success
Propensity Score Matching	-0.713** (0.263)	0.065+ (0.035)
Nearest Neighbor Matching	-0.631*** (0.175)	0.080** (0.030)

Note: The Abadie-Imbens standard errors are in parentheses. For PSM, a caliper of 0.01 (maximum propensity score distance) is applied, dropping up to 40 observations; for NNM, a caliper of 5 (maximum Mahalanobis distance) is applied, dropping up to 24 observations. *** p<0.001, ** p<0.01, * p<0.05, + p<0.1

B-4: Alternative Estimators

We also examine whether the main market-clearing results are sensitive to estimator choice. In the main analyses, *TimeToAccept* is estimated using a logged linear model, and *ProjectSuccess* is estimated using a linear probability model. Although these specifications are straightforward to interpret, alternative estimators may be more appropriate for the distributional properties of the dependent variables.

For *TimeToAccept*, we estimate a Cox proportional hazards model, treating contract award as a time-to-event outcome (Cox 1972). A positive coefficient or hazard ratio greater than one indicates that projects are awarded more quickly after the visibility change. For *ProjectSuccess*, we estimate random-effects Logit and Probit models to account for the binary nature of the dependent variable.

Table B4 reports the results. The Cox model shows that the post-change indicator is positive and statistically significant, and the corresponding hazard ratio is greater than one. This result indicates that sealed bidding is associated with a higher rate of contract award, consistent with faster market clearing. The random-effects Logit and Probit models also show positive and statistically significant effects on *ProjectSuccess*. Together, these alternative specifications confirm that the main conclusions are not artifacts of the baseline linear-model choices.

Table B4. Effect of Visibility Change on Market-Clearing Efficiency

Dep. Variable	Time to Accept		Project Success	
	Cox: Coef.	Cox: Hazard Ratio	RE-Logit	RE-Probit
AfterChange	0.360**	1.433**	0.367*	0.218*
	(0.115)	(0.165)	(0.163)	(0.096)
EmployerExperience	0.315***	1.370***	0.155	0.096
	(0.090)	(0.123)	(0.230)	(0.135)
MaxBid	0.027	1.027	0.063*	0.037*
	(0.017)	(0.017)	(0.027)	(0.016)
DescriptionLength	-0.064+	0.938+	0.099	0.059
	(0.035)	(0.033)	(0.066)	(0.039)
ProjectDuration	-0.736***	0.479***	-0.671***	-0.401***
	(0.066)	(0.032)	(0.091)	(0.053)
Observations	802		1,926	1,926
Number of Employers	423		967	967
Log-Likelihood				
Employer Region Dummies	YES		YES	YES
Weekday Dummies	YES		YES	YES
Project Type Dummies	YES		YES	YES

Note: Robust standard errors are in parentheses. *** p<0.001, ** p<0.01, * p<0.05

B-5: Heterogeneous Effects

Finally, we examine whether the effect of the visibility change differs across market participants and project contexts. These analyses are theoretically relevant because the value of bid visibility should depend on the experience of employers and workers. Experienced employers may be better able to evaluate incoming offers quickly and may therefore be less affected by changes in bid visibility. Experienced workers may also rely less on visible competitor information because they have accumulated knowledge about pricing, project fit, and employer preferences.

Table B5a examines the moderating role of employer experience. The results show that the interaction between *AfterChange* and *EmployerExperience* is positive for *TimeToAccept* and negative for *ProjectSuccess*. These patterns suggest that the market-clearing benefits of sealed bidding are weaker for more experienced employers. One interpretation is that experienced employers already know how to evaluate candidates efficiently, so removing bid visibility produces smaller improvements in their contracting speed and success probability.

Table B5b examines the moderating role of average worker experience. The interaction between *AfterChange* and *MeanWorkerExperience* is positive and statistically significant for *TimeToAccept*, indicating that the reduction in time to contract award is weaker when the participating worker pool is more experienced. This finding is consistent with the argument that experienced workers rely less on visible competing bids and therefore adjust their timing behavior less when bid visibility is removed. The interaction is not statistically significant for *ProjectSuccess*, suggesting that worker experience moderates the speed of market clearing more strongly than the probability of contract formation.

Overall, the heterogeneity analyses suggest that the effect of bid visibility is not uniform across market conditions. The benefits of sealed bidding appear stronger in settings where employers or workers are less experienced and therefore more likely to be affected by the information environment.

Table B5a. Effect of Visibility Change: Moderating Role of Employer Experience

Dep. Variable	Time to Accept	Project Success
AfterChange	-0.587***	0.085**
	(0.160)	(0.027)
AfterChange × EmployerExperience	0.445*	-0.217***

	(0.187)	(0.047)
EmployerExperience	-0.782***	0.284***
	(0.211)	(0.070)
MaxBid	-0.052*	0.012*
	(0.027)	(0.005)
DescriptionLength	0.141**	0.017
	(0.050)	(0.012)
ProjectDuration	1.107***	-0.121***
	(0.096)	(0.015)
Constant	-0.053	0.375**
	(0.371)	(0.119)
Observations	802	1,926
Number of Employers	423	967
Adjusted R^2	0.310	0.0958
Employer Region Dummies	YES	YES
Weekday Dummies	YES	YES
Project Type Dummies	YES	YES

Note: Robust standard errors are in parentheses. *** p<0.001, ** p<0.01, * p<0.05

Table B5b. Effect of Visibility Change: Moderating Role of Mean Worker Experience

Dep. Variable	Time to Accept	Project Success
AfterChange	-0.793***	0.018
	(0.210)	(0.033)
AfterChange × MeanWorkerExperience	1.702***	-0.100
	(0.326)	(0.083)
MeanWorkerExperience	-2.129***	0.297***
	(0.296)	(0.084)
EmployerExperience	-0.300***	0.030
	(0.083)	(0.039)
MaxBid	-0.034	0.006
	(0.025)	(0.005)
DescriptionLength	0.107*	0.027*
	(0.048)	(0.011)
ProjectDuration	1.074***	-0.104***
	(0.098)	(0.015)
Constant	0.803+	0.186
	(0.411)	(0.119)
Observations	802	1,926
Number of Employers	423	967
Adjusted R^2	0.355	0.138
Employer Region Dummies	YES	YES
Weekday Dummies	YES	YES
Project Type Dummies	YES	YES

Note: Robust standard errors are in parentheses. *** p<0.001, ** p<0.01, * p<0.05

E-Companion C: Participation and Competition, and Match Quality

C-1: Early Arrival of Bidders

The main analysis shows that workers submit bids earlier after the platform transitions to sealed bidding and that projects clear more quickly. To provide additional evidence on the operating process linking these results, we examine whether sealed bidding changes the early arrival of bidders at the project level. This analysis complements the bid-level timing analysis in Section 5 by asking whether employers receive timely offers early in the project procurement process.

We construct three project-level measures of early bidder arrival. First, *HasBidWithin24h* is an indicator equal to one if the project receives at least one bid within the first 24 hours after posting. Second, *LogNumberofBiddersWithin24h* measures the logged number of bidders who submit bids within the first 24 hours. Third, *EarlyBidderShareWithin24h* measures the share of all participating bidders who submit bids within the first 24 hours. We also construct analogous measures for experienced bidders, defined as workers with prior completed jobs at the time of bidding. These experienced-bidder measures provide a conservative proxy for the early arrival of more actionable offers, because employers may be able to evaluate workers with prior platform experience more readily than workers with no prior completed jobs.

We estimate project-level regressions using the same project and employer controls as the baseline market-clearing specifications, including employer experience, maximum bid, description length, project duration, weekday dummies, employer-region dummies, and project-type dummies. These analyses should be interpreted as supplemental process evidence rather than as separate causal mediation tests.

Table C1 reports the results. Column 1 shows that sealed bidding is associated with a higher probability that a project receives at least one bid within the first 24 hours. Column 2 shows that sealed bidding does not significantly reduce the number of bidders arriving within the first 24 hours. Column 3 shows that sealed bidding is associated with a larger share of bidders arriving within the first 24 hours. Columns 4 through 6 show even stronger patterns for experienced bidders: sealed bidding significantly increases the probability that a project receives at least one experienced-bidder offer within 24 hours, increases the number of experienced bidders arriving within 24 hours, and increases the share of experienced bidders arriving within this early window.

These results indicate that sealed bidding changes the timing and composition of worker participation. Although sealed bidding reduces overall bidder participation in the main analysis, it does not reduce the number of bidders arriving early and even increases the number of experienced bidders arriving within the first 24 hours. Thus, fewer observed bidders do not necessarily imply that employers receive fewer timely or actionable offers. Instead, the evidence suggests that sealed bidding accelerates the arrival of early offers, especially from experienced workers, which helps explain why employers can award contracts faster and why projects are more likely to clear after the visibility change.

Table C1. Effect of Visibility Change on Early Arrival of Bidders

Dep. Variable	Has Bid Within 24h	Log Number of Bidders within 24h	Early Bidder Share within 24h	Has Experienced Bid Within 24h	Log Number of Experienced Bidders within 24h	Early Experienced Bidder Share within 24h
AfterChange	0.035*	0.031	0.091***	0.193***	0.284***	0.154***
	(0.016)	(0.047)	(0.017)	(0.027)	(0.040)	(0.015)
EmployerExperience	-0.015	-0.124***	0.035*	0.019	-0.010	0.038*
	(0.011)	(0.036)	(0.016)	(0.018)	(0.027)	(0.018)
MaxBid	0.002	-0.010	0.004	0.004	0.000	0.008*
	(0.003)	(0.008)	(0.003)	(0.005)	(0.007)	(0.004)
DescriptionLength	-0.000	0.106***	-0.025***	0.013	0.059***	-0.026**
	(0.005)	(0.017)	(0.007)	(0.010)	(0.015)	(0.008)
ProjectDuration	-0.043***	-0.145***	-0.170***	-0.086***	-0.130***	-0.080***
	(0.009)	(0.026)	(0.010)	(0.015)	(0.022)	(0.008)
Observations	1,926	1,926	1,926	1,926	1,926	1,926
Number of Employers	967	967	967	967	967	967
Adjusted R2	0.031	0.088	0.213	0.057	0.074	0.166
Employer Region Dummies	YES	YES	YES	YES	YES	YES
Weekday Dummies	YES	YES	YES	YES	YES	YES
Project Type Dummies	YES	YES	YES	YES	YES	YES

Note: Standard errors are in parentheses. + p<0.10, *** p<0.001, ** p<0.01, * p<0.05

C-2: Repeated Bidding Activities

In the main paper, we show that sealed bidding improves market-clearing efficiency by reducing time to contract award and increasing the probability of contract formation. We also examine whether these improvements are accompanied by changes in the competitive structure of the bidding process. This section provides additional evidence on repeated bidding activity.

If open bidding allows workers to observe competitors and adjust their behavior over time, workers may engage in more monitoring and bid revision. By contrast, sealed bidding reduces the informational benefit of monitoring competitors after bids are submitted. Therefore, we expect sealed bidding to reduce repeated bidding activity.

Figure C-1 provides model-free evidence on the number of bids and the number of bids per bidder before and after the visibility change. Table C2 reports regression results using two dependent variables: the total number of bids submitted to a project and the number of bids submitted per bidder. The coefficient on *AfterChange* is negative and statistically significant in both specifications. These results indicate that sealed bidding is associated with fewer total bids and fewer bids per bidder.

These findings are consistent with the proposed mechanism. When competing bids are hidden, workers have fewer opportunities to monitor and revise their bids in response to competitors. As a result, bidding activity becomes less repetitive. Together with the main results on participation and average bid amounts, these findings suggest that sealed bidding restructures competition: fewer workers participate and submit fewer repeated bids, but participating workers submit more competitive offers.

Figure C-1. Comparing Additional Bid Information

(a). Number of Bids and Regime Change **(b). Number of Bids Per Bidder and Regime Change**

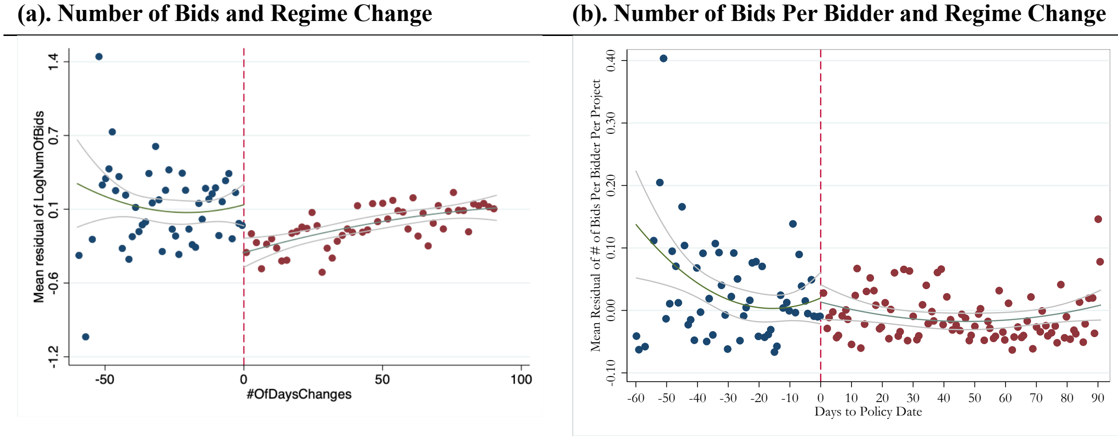

Table C2. Effects of Visibility Change on Additional Bid Competition

Dep. Variables	Number of Bids	Number of Bids Per Bidder
AfterChange	-0.1779***	-0.0356***
	(0.0470)	(0.0098)
EmployerExperience	-0.1283***	0.0028
	(0.0309)	(0.0062)
MaxBid	-0.0113	0.0008
	(0.0091)	(0.0017)
DescriptionLength	0.1312***	0.0049
	(0.0202)	(0.0038)
ProjectDuration	0.1333***	0.0080
	(0.0283)	(0.0056)
Observations	1,926	1,926
Adjusted R^2	0.153	0.024
Number of Employers	967	967
Employer Region Dummies	YES	YES
Weekday Dummies	YES	YES
Project Type Fixed Effects	YES	YES

Note: Robust standard errors are in parentheses. *** p<0.001, ** p<0.01, * p<0.05, +p<0.1

C-3: Effects of Visibility Change on Match Quality

The main paper uses employer ratings as the primary measure of post-award match quality. Because *WorkerRating* is bounded and concentrated near the upper end of the rating scale, we conduct additional robustness checks using ordered response models. These models account for the ordinal nature of employer ratings and provide an alternative to the linear specification reported in the main paper.

Table C3 reports ordered Logit and ordered Probit estimates. Across the ordered specifications, the coefficient on *AfterChange* remains positive and is statistically significant in most specifications. These findings suggest that the positive association between sealed bidding and employer-rated match quality is not merely driven by the use of a linear model.

Overall, the ordered-model results provide additional support for Hypothesis 4. Contracts formed after the transition to sealed bidding are associated with higher employer ratings, suggesting that faster market clearing does not come at the expense of observed post-award match quality.

Table C3. Effects of Visibility Change on Matching Quality: Robustness Checks

Dep. Variable	Worker Rating					
	Ordered Logit			Ordered Probit		
AfterChange	0.571**	0.567**	0.416	0.411***	0.404***	0.320**
	(0.257)	(0.257)	(0.265)	(0.136)	(0.136)	(0.142)
DescriptionLength		-0.0143	0.0109		-0.00120	0.0142
		(0.106)	(0.106)		(0.0541)	(0.0534)
EmployerExperience			0.209			0.119*
			(0.129)			(0.0624)
WorkerExperience			0.235**			0.128**
			(0.117)			(0.0605)
SameCountry			0.143			0.0788
			(0.248)			(0.130)
WinningBid		-0.108	-0.122		-0.0410	-0.0485
		(0.0843)	(0.0850)		(0.0442)	(0.0446)
Observations	802	802	802	802	802	802
Project-type Fixed Effects	No	YES	YES	No	YES	YES

Note: Robust standard errors are in parentheses. *** p<0.001, ** p<0.01, * p<0.05, +p<0.1

C-4: Rehire as an Additional Measure of Match Quality

We also examine *Rehire* as an additional measure of match quality. *Rehire* equals one if the employer subsequently hires the same worker for a future project. Compared with attitude-based employer ratings, rehiring captures a more behavior-based form of post-award satisfaction because it indicates that the employer is willing to work with the same worker again.

Table C4 reports the results. The coefficient on *AfterChange* is positive and statistically significant in the specifications with fewer controls. However, after adding the full set of project, employer, and worker characteristics, the coefficient becomes smaller and statistically insignificant. These results suggest that contracts formed after the visibility change may be associated with a greater likelihood of future rehiring, but the evidence is sensitive to model specification.

Accordingly, we interpret the rehire result as suggestive rather than conclusive evidence. The more robust evidence for match quality comes from employer ratings, while the rehire outcome analysis provides complementary evidence that does not contradict the main interpretation.

Table C4. Effects of Visibility Change on Rehire

Dep. Variable		Rehire	
AfterChange	0.139***	0.124***	0.0437
	(0.0353)	(0.0357)	(0.0359)
DescriptionLength		-0.0193	-0.00370
		(0.0147)	(0.0126)
EmployerExperience			0.0986***
			(0.0179)
WorkerExperience			0.0938***
			(0.0194)
SameCountry			-0.0370
			(0.0319)
WinningBid		0.0179	0.00897
		(0.0147)	(0.0140)
Observations	802	802	802
Adjusted R^2	0.015	0.027	0.091
Project-type Dummies	No	YES	YES

Note: Robust standard errors are in parentheses. *** $p<0.001$, ** $p<0.01$, * $p<0.05$

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